MILITARY SPECIFICATION
CRANE, HYDRAULIC, WHEEL MOUNTED, 7-1/2 AND 15 TON CAPACITIES

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This Specification covers diesel engine powered, pneumatic tired, full revolving, hydraulically operated cranes, 7-1/2 and 15 ton capacities with attachments.

1.2 Classification. Cranes shall be of the following sizes, as specified (see 6.2):

Size A	7-1/2 Ton Capacity
Size B	15 Ton Capacity

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	Fuel Oil, Diesel
W-B-131	Battery, Storage: Vehicular, Ignition Lighting and Starting

MILITARY

MIL-P-514	Plates, Identification, Instruction, and Marking, Blank
MIL-T-704	Treatment and Painting of Material
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRS, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-H-83282 Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537

STANDARDS

FEDERAL

FED-STD-595 Color

MILITARY

MIL-STD-12	Abbreviation for Use on Drawings, and in Specification, Standards, and Technical Documents
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standard Order of Precedence, for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-ST0-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-808	Finishes, Protective, and Codes, for Finishing
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Nontactical Wheeled Vehicles Treatment, Painting, Identification Marring and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MS 51117	Pintle Assembly, Towing: 100,000 Pound Capacity

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date specified in the solicitation.

US DEPARTMENT OF AGRICULTURE (USDA)

5100-1A Spark Arrestors for Internal Combustion Engines

(Application for copies should be addressed to US Department of Agriculture, Forest Service, Equipment Development Center, Dan Dimas CA 91773.)

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC (ANSI)

A92.5	Boom Supported Elevated Work Platforms
B30.5	Mobile and Locomotive Cranes

(Application for copies should be addressed to the American National Standard Institute, Incorporated, 1430 Broadway, New York NY 10017.)

POWER CRANE AND SHOVEL ASSOCIATION (PCSA)

PCSA No. 4	Mobile Power Crane and Excavator and Hydraulic Crane Standards

(Application for copies should be addressed to Construction Industry Manufacturers Association, Marine Plaza, Suite 1700, 111 E. Wisconsin Avenue, Milwaukee Wisconsin 53202.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE STANDARDS AND RECOMMENDED PRACTICES

J88	Sound Measurement - Earthmoving Machinery - Exterior
J375	Radius of Load and Boom Angle Measurement System
J376	Load Indicating Devices in Lifting Crane Service
J534	Lubrication Fittings
J765	Crane Load Stability Test Code
J820	Crane Hoist Line Speed and Power Test Code
J899	Operator's Seat Dimensions - Construction and Industrial Equipment Design
J919	Sound Measurement - Earthmoving Machinery - Operator - Singular Type
J994	Alarm - Back Up
J1063	Cantilevered Boom Crane Structures - Method of Test
J1152	Minimum Performance Criteria for Braking Systems for Off Highway, Rubber Tired, Self-Propelled Construction Machines
J1180	Telescopic Boom Length Indicating System
J1349	Engine Power Test Code - Spark Ignition and Diesel

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Yearbook

(Application for copies should be addressed to the Tire and Rim Association Incorporated, 3200 West Market Street, Akron OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities, and services not specified herein shall be selected in accordance with MIL-STD-143.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced specification shall be suitable in every respect for this type of equipment. Wood shall not be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design and construction. The crane shall be constructed so that parts shall not work loose in service, and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the crane shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so there is no adverse interference with each other or with the operation and shall be readily accessible for maintenance, operation, removal and replacement. The crane shall comply with all requirements of SAE J765, SAE J1063 and ANSI B30.5.

3.1.3.1 Reliability. The crane, including all components, shall have an observed Mean-Time-Between-Failure (MTBF) of not less than 100 hours. The crane shall also have an operational availability of not less than 0.99. A failure is defined as the event or inoperable state in which any item or part of an item does not, or would not, perform as previously specified. Operational availability (Aø) is defined as the percent of time that the system is capable of satisfactorily performing in the operational environment. Mathematically, Aø is expressed as:

$$A_ø = MTBF/(MTBF + MRT),$$

where MRT is Mean Repair Time.

Mean repair time (MRT) is the elapsed time to accomplish all repairs divided by the number of repairs performed. Mathematically, MRT is expressed as:

$$MRT = T/N,$$

where T is Elapsed Time
N is Number of Repairs.

The elapsed time includes all maintenance actions required to correct the malfunction, including time to prepare, remove, replace, repair, adjust, and check. MRT does not include administrative or supply delays. Note that time to troubleshoot is not included; thus, this is not the standard definition for MRT.

Definitions of reliability terms not included herein shall be as found in MIL-STD-721 and MIL-STD-781. In the event of a conflict between these documents, MIL-STD-781 shall take precedence. The crane shall be designed for a life expectancy of at least 15 years.

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3.1.3.2 Maintainability. The crane shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the crane. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing, and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.
e. Maintenance with a minimum number of tools.

f. Maximum repair time, excluding time to troubleshoot, of no greater than eight hours.

g. Maximum time to perform all daily scheduled maintenance and servicing tasks of no greater than 15 minutes.

3.1.3.2.1 Maintenance tasks. The crane shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

a. Remove, replace, and adjust all engine driven belts - 3/4 hour.

b. Remove and replace alternator - 1/2 hour.

c. Remove and replace all filters, screens, and strainers in all hydraulic systems - one hour.

d. Remove and replace engine coolant system hoses - 1/2 hour.

e. Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase - one hour.

f. Remove and replace fuel filter element(s) - 1/2 hour.

g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1/2 hour.

h. Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers, and refill converter and transmission - one hour.

i. Remove and replace starter - one hour.

j. Remove or install the engine/transmission package, two mechanics - four hours.

k. Reeve hook block - 1/2 hour (begin with one part of line and reeve five parts of line).

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an unsymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be located so they are readily accessible and do not require special tools for proper servicing.

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3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings conforming to SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fitting shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of inaccessibility. The crane shall be designed and constructed to operate satisfactorily when lubricated with the standard military lubricants specified in MIL-M-8090, except in Table II where hydraulic system shall be in accordance with MIL-H-83282.

3.1.3.7 Protective features. All space in which work is performed during operation, service and maintenance of the unit shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.8 Component protection. All oil, hydraulic and air system tubing and all electrical lines shall be securely clipped to the frame or the body structure and, except where a through frame connector is necessary, shall be furnished with metal protective looms or grommets at each point where they pass through a panel or structural member.

3.1.3.9 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the crane.

3.1.3.10 Bushings and bearings. Bushings or bearings, as appropriate, shall be provided at any location where there is relative rotation between two parts, except hinges and cables.

3.1.4 Environmental conditions. The crane shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full time exposure to temperatures ranging from -20°F (-40°F see 6.2), to +125°F.

b. Full time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm falling at a rate of 13 cm per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80cm^2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1,750 ft/min, concentration 0.5 g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5,700 ft/min, concentration 0.0773 g/ft^3.

3.1.5 Stability. With lubricating and cooling systems full, fuel tank(s) at least half full with crane resting on firm, level, and a uniform supporting surface, with outriggers unsupported, with the boom fully retracted and in travel position; the weight on the front axle shall be within 40 to 60 percent of the total weight of the crane and shall conform to the following additional requirements:

a. With the longitudinal axis of the rotating superstructure of the crane at 90° to the longitudinal axis of the chassis, the boom retracted and set at the minimum recommended radius, the total load on all

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wheels on the side of the chassis under the boom shall not be less than 15 percent of the total weight of the crane.

b. With the longitudinal axis of the rotating superstructure of the crane in line with the longitudinal axis of the carrier, in either direction, the boom retracted and set at the minimum recommended radius, the total load on all wheels under the lighter loaded end of the chassis shall not be less than 15 percent of the total weight of the crane.

3.1.6 Electromagnetic interference. The crane shall comply with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference (emission and susceptibility), except that the limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.1.7 Air transportability. Size A crane shall be air transportable without shoring in accordance with MIL-STD-1791 in Types C-130, C-141, and C-5 Aircraft and shall be air transportable in a Type C-17 Aircraft. Size B crane shall be air transportable without shoring in accordance with MIL-STD-1791 in types C-5 and C-17. Preliminary design guidelines for the Type C-17 are 142 inches maximum height, 204 inches maximum width, 812 inches maximum length, 9° ramp angle, and floor load capacity equal to that allowed for the Type C-5 Aircraft.

3.1.8 Nuclear safety. When specified (see 6.2), the crane shall comply with the following requirements for lifting nuclear loads:

1. Controls shall ensure that the load is under positive operator control.

2. The crane shall have features that ensure prompt and automatic stop (except for cable swing) in case of operating mechanism and/or hydraulic failure (see 3.4.2).

3. The design of the crane shall permit completely synchronized operations.

4. The crane shall have features that will automatically neutralize controls in the absence of a manual force.

5. The reduced lift capacity, positioning limits and rates of the crane shall be identified on separate load charts, based on the following criteria:

a. The crane shall have a minimum safety factor of three for all loading sustaining (basic structure) parts, based on the material minimum yield strength.

b. The crane shall have a minimum safety factor of ten for blocks and rope falls, fiber rope and webbing, based on the material minimum ultimate strength.

c. Accessory parts (such as hooks, rings, shackles, slings and wire rope) shall have a safety factor of five based on ultimate strength.

6. Hooks shall be equipped with throat opening devices.

7. The crane shall have stops or limit switches to stop the wire rope when the hook reaches its upper limit.

8. No reference to nuclear certification shall be made on the crane markings or in the technical data.

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3.2 Performance.

3.2.1 Rated capacity. The crane shall have a rated capacity as specified at the designated radii. The capacity on outriggers and on rubber shall be for 360° rotation. Pick and carry capacity shall be with the boom centered over the front and shall be at least equal to the specified capacities on rubber. The crane shall comply with the requirements of SAE J765 for stability and SAE J1063 for structural integrity at the specified loads. The weight of the hook and block shall be in addition to the rated capacities listed below:

MINIMUM RATED CAPACITY
(POUNDS)

	SIZE A		SIZE B	
RADII	ON RUBBER	ON OUTRIGGERS	ON RUBBER	ON OUTRIGGERS
10	7,000	15,000	18,400	30,000
15	5,000	11,500	9,500	24,500
20	3,000	6,750	5,500	17,400
25	1,700	4,500	3,200	13,900
30	1,100	3,775	2,000	11,200
35	650	2,300	1,000	8,600
40				6,900
45				5,300
50				4,100
55				3,600
60				2,100

3.2.2 Mobility.

3.2.2.1 Travel speed. The unloaded crane shall be capable of a forward speed of at least 25 MPH and a reverse speed of at least 4 MPH.

3.2.2.2 Pick and carry speed. The crane shall be capable of carrying the rated pick and carry load at a forward or reverse speed of at least 2 MPH.

3.2.2.3 Cross country operation. The unloaded crane shall be capable of operating over cross country terrain at an average speed of at least 10 MPH.

3.2.2.4 Turning radius. The turning radius for the Size A crane shall be no greater than 234 inches, for the Size B, no greater than 240 inches.

3.2.3 Hoist. For the Size A crane, the winch shall have a minimum of 7,500 pounds line pull at a minimum line speed of 150 FPM on the first layer. For the Size B crane, the winch shall have a minimum of two speeds. Low speed shall have a minimum of 10,500 pounds line pull at a minimum line speed of 230 FPM on the first layer, at a line speed of not less than 400 FPM. The available line pull and speed shall be as rated in accordance with PCSA Standard Number 4, for the operational requirements of the crane hoisting drums. Hydraulically controlled power load lowering shall be provided. For both the Size A and Size B cranes, the power down shall have sufficient capacity to hold a load without slipping on a single part of line, equal to or greater than the permissible line pull for the winch. The power load lowering shall provide positive load control.

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3.2.4 Boom rotation speed. The boom shall have a minimum rotation speed of two revolutions per minute.

3.2.5 Superstructure rotation. The power transmission system used to rotate the superstructure shall be a hydraulic mechanical arrangement with a continuous swing rotation angle of not less than 360°. Swing speed within this range shall be smooth and constant, and adequate control shall be provided in the hydraulic circuit to ensure smooth starting and stopping. The crane shall be capable of simultaneously lifting and rotating, throughout 360°, a load equal in pounds to the line pull determined from the condition as indicated herein. The swing mechanism shall be equipped with a swing brake (parking and dynamic) and swing lock in accordance with PCSA Number 4.

3.3 Carrier.

3.3.1 Frame. The frame assembly shall provide a rigid unit structure and shall be of bolted, riveted and/or welded steel construction.

3.3.2 Powertrain.

3.3.2.1 Engine. The crane shall be equipped with an industrial type liquid cooled diesel engine of not less than four cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel in accordance with VV-F-800 and shall operate satisfactorily on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624 and Grade JP-8 in accordance with MIL-T-83133. A power loss of 10 percent (maximum) for JP-4 or JP-5 shall be permitted. Any engine that is adversely affected by any of the above fuels is not acceptable.

3.3.2.1.1 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the engine manufacturer's recommended maximum or 280°F (whichever is lower). Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided.

3.3.2.1.2 Engine starting system. The engine shall start within 15 seconds in any ambient temperature within the required operating range of the crane. Installed glow plugs, fluid starting aids and heat from the winterization system (see 3.3.7) may be used prior to and during the start period to facilitate engine starting under the following conditions:

TEMPERATURE RANGE (DEGREES F)	STARTING AID PERMITTED
40 through 125	None
1 through 39	Glow plugs and fluid starting aids
-40 through 0	Glow plugs, fluid starting aids, and heat from the winterization system

The starting system shall be equipped with an interlock device to prevent starter engagement when the transmission is in any position other than neutral.

3.3.2.1.3 Exhaust system. The exhaust system shall incorporate a spark arrestor muffler in accordance with USDA Standard 5100-1, EQUIPPED with a cleanout plug of at least 3/4 inch. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall not be directed toward the ground. The exhaust system, including all fasteners, shall be

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constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain.

3.3.2.2 Transmission. The crane shall be equipped with a torque converter and a power shift or fully automatic transmission providing not less than three forward and two reverse gear ratios. The transmission shall be compatible with the engine and drive line components. The transmission shall have a maximum input rating greater than the maximum net torque and power output of the engine.

3.3.2.3 Axles. The front and rear axles shall be of sufficient capacity to sustain the imposed loads and torques imposed by all operations specified herein. Maximum practical component interchangeability shall be maintained between the front and rear axles. The rear axle shall be of the oscillating type.

3.3.3 Steering. The crane shall be equipped with an automotive type steering wheel to actuate a hydraulic power assisted or fully hydraulic steering system. The steering system shall provide two wheel steer, four wheel steer, and oblique steer modes. Provisions shall be made for adjustment to compensate for component wear. Lubrication fittings shall be provided at all friction points. Steering for the front wheels shall be designed to maintain manual steering in case of engine failure.

3.3.3.1 Steering wheel. The steering wheel shall have a diameter of 18 inches minimum. Not less than four inches clearance shall be provided around the steering wheel in all directions.

3.3.4 Brakes.

3.3.4.1 Service brakes. Hydraulic or air actuated disc or drum type service brakes shall be provided. The service brakes shall hold the crane on a 25 percent slope with the maximum pick-carry load. The maximum allowable operator force required to actuate braking system (service and parking brake) shall not exceed 100 pounds of foot pressure. The crane shall be capable of an emergency stop due to engine or hydraulic system failure in 32 feet. The service brakes shall be automatic self-adjusting type to comply with SAE J1152.

3.3.4.2 Parking brakes. Mechanical or spring set parking brake shall be provided. The parking brakes shall comply with SAE J1152. The parking brakes shall be capable of holding the crane with the pick and carry load on a longitudinal slope of not less than 15 percent without slippage.

3.3.5 Wheel and tires. The crane shall be equipped with commercial tires and wheels which meet Tire and Rim Association requirements for this application. If tube type tires are provided, inner tubes shall comply with Tire and Rim Association requirements.

3.3.6 Batteries. The crane shall be equipped with at least two 12 volt batteries, rated at a total of 1000 cold cranking amps (CCA) minimum. The batteries shall comply with the maintenance free characteristics of W-B-131. The batteries shall be mounted in an acid resistant tray; a sliding carriage shall be provided if the batteries must be moved sideways to permit access for service or replacement. A master switch shall be provided to disconnect the batteries from the electrical system.

3.3.7 Winterization system. When specified (see 6.2), the crane shall be equipped with a winterization system to permit storage and operation in ambient temperatures as low as -40°F. Heaters shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding the crane. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the crane end and a standard weatherproof three pronged (two power, plus one ground) male connector at the other end. Electrical apparatus shall conform to FMCSR 393.77(c) (7). Electrical insulation of connective cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with

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crane component operation and without loose excess wire. Provisions for storage of the cable shall be provided in the crane cab. Heaters shall be furnished as follows:

a. A coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose, controlled by a thermostat with an operating range of 140°F to 160°F.

b. An immersion type engine oil heater, 300 watt minimum rating, shall be installed in oil pan through any convenient opening. The oil heater shall be controlled by the coolant heater thermostat.

c. A battery heater shall be provided having a capacity adequate to maintain battery electrolyte at a temperature of not less than 10°F during crane exposure in ambient air temperatures as low as -40°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

d. A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°F to an outlet temperature of 10°F, with a flow rate not less than the maximum fuel demand of the engine fuel system.

3.3.8 Fuel system.

3.3.8.1 Fuel tank. A fuel tank shall be provided to allow the crane to operate 8 hours (minimum) under any condition. The tank shall have a fill opening of three inches minimum readily accessible to personnel standing on the ground. It shall be located and mounted so as to provide maximum protection from damage.

3.3.8.2 Fuel filter. At least one primary filter/strainer of the fuel water separator type shall be located in the fuel line between the fuel tank and the fuel pump. At least one secondary filter for filtering fine dirt particles shall be located on the pressure (outlet) side of the fuel pump. All fuel filters shall be installed so that the elements can be readily serviced without disturbing other items. If located below the fuel tank level, a shutoff valve shall be provided between it and the tank so that the element may be changed without draining the tank.

3.4 Winch and boom.

3.4.1 Winch. The hoist winch shall be hydraulic motor driven and shall incorporate an automatic brake capable of holding the maximum rated load. The winch shall be equipped with an automatic cable tensioning device or equivalent, to ensure sufficient pressure maintained on the cable during light/no load reel in operation. This device shall not reduce cable life by more than 5 percent.

3.4.1.1 Drum. The drum shall be designed to ensure proper spooling of the rope and shall incorporate flanges that extend at least one inch over the top layer of rope at all times. The drum diameter shall be in accordance with ANSI B30.5. The drum shall be weatherproof.

3.4.1.2 Wire rope. The load hoist rope shall be not less than 6 X 19 improved plow steel independent wire rope center having a strength factor of not less than five. Minimum rope diameter shall be 0.50 inches for the Size A crane and 0.62 inches for the Size B crane. Sufficient rope shall be provided to allow performance of all lifts specified herein.

3.4.1.3 Hook and block. The hook and block shall be of sufficient weight to ensure payout of the rope with no load on the hook. The hook shall be of the swivel type, with a minimum factor of safety of five, based on the material minimum ultimate strength. The hook shall be equipped with a throat opening device. The block assembly shall provide for attachment of rope for not less than two part purchase.

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3.4.1.4 Anti-two block system. An anti-two block switch shall be provided to prevent rope breakage and wedging or binding of the hook when it is at the boom end. Sheaves used in multiple rope reeving shall have a pitch diameter not less than 16 times the nominal diameter of the rope.

3.4.2 Boom. The boom shall be of the hydraulically operated telescoping type, consisting of a base section and sliding sections, capable of extension and retraction to the specified lengths. The boom tail shall not project more than 23 inches beyond the front of the carrier or more than nine inches beyond the rear of the carrier in any operating condition. A boom angle indicator of the electronic type shall be provided. The pointer shall be clearly visible to the operator. Upon hydraulic failure, the boom shall maintain angle and not permit uncontrolled lowering of the boom in accordance with ANSI B30.5.

	SIZE A	SIZE B
Minimum Extended Boom Length	40 feet	65 feet
Maximum Retracted Boom Length	21 feet	36 feet

3.4.3 Hydraulic system. The hydraulic system shall include the pump, cylinders, reservoir, filter(s) (spin on cartridge type), pressure relief valve and all necessary lines, restrictors and control valves to ensure positive control of the winch, outriggers and boom in all operational conditions and in the event of hydraulic pressure loss. The pump shall be driven by the crane engine and shall have sufficient capacity to energize all hydraulically powered components, except steering and outriggers, operating simultaneously. Provision shall be made for bleeding all air from the system. All hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure and a proof pressure of two times the maximum operating pressure. Hydraulic reservoir shall incorporate a manual shutoff valve to prevent fluid loss during maintenance and a sight gauge or dipstick to measure the fluid level of hydraulic system. The reservoir shall include a filler opening of 2 1/4 inches (minimum) with a filter screen and captive cap.

3.5 Operator station. The operator station for the Size A crane shall be mounted to the carrier or the superstructure. The operator station for the Size B crane shall be mounted to the superstructure on the left side of the boom facing forward so as to rotate with the boom.

3.5.1 Controls. All controls and indicators shall be located within easy reach of the operator while sitting in the operator's seat. Controls shall ensure that the load is under positive operator control; controls shall be designed so that they are automatically neutralized in the absence of an external force.

3.5.2 Instruments. The following instruments shall be installed on the instrument panel so that they are clearly visible to the operator in the cab:

a. Tachometer.

b. Engine oil pressure gauge.

c. Engine coolant temperature gauge.

d. Transmission/torque converter oil temperature gauge.

e. Hydraulic oil temperature gauge.

f. Fuel level gauge.

g. Voltmeter or ammeter.

h. Hourmeter.

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i. A red warning light mounted onto the dash indicating when the winterization system is operating.

Symbolic markings shall be either embossed or depressed on or near the respective control or gauge. Instrumentation and electrical controls shall have a quick disconnect plug on the wiring harness to allow removal of the complete panel or sections of panels for maintenance.

3.5.3 Seat. A heavy duty black vinyl covered seat in accordance with SAE J899 shall be provided. The seat shall have a minimum of three inches vertical adjustment. A seat belt shall be provided.

3.5.4 Cab. Unless otherwise specified (see 6.2), a fully enclosed cab shall be provided. The cab shall be of sheet steel construction and shall have tinted safety glass on all four sides and in the roof for good visibility. A cab heater and means for defrosting the windshield and rear window shall be provided. The cab door latch shall be sufficient to prevent the door from accidently opening and shall be operable by a person wearing heavy arctic mittens. An electric windshield wiper, dome light and horn shall be supplied.

3.6 Outriggers. Four hydraulically operated outriggers shall be provided, operable separately or in any combination from the operator station. The outriggers shall be capable of extending not less than six inches below ground level. Minimum spread from center to center of the pads shall be 152 inches for the Size A crane and 196 inches for the Size B crane. The outrigger pad size shall be such that the maximum pad force does not exceed 150 psi ground pressure.

3.7 Lights. The crane shall be equipped with two sealed beam headlights with provisions for vertical and horizontal adjustment, two tail lights, brake lights, front and rear directional signal lights, and side marker lights at each corner. The cranes shall be equipped with a minimum of two boom lights to illuminate load, two work lights on the cab to illuminate work area and one spotlight capable of 360° rotation and 180° elevation. All lights shall be mounted in protected locations.

3.8 Accessories.

3.8.1 Fire Extinguisher. One Halon 1211 Fire Extinguisher in accordance with A-A-1108. It shall be of the type and size in accordance with ANSI B30.5.

3.8.2 Back-up alarm. An audible alarm shall be provided to caution personnel in the area when the crane is in reverse gear. The alarm shall be in accordance with SAE J994.

3.8.3 Pintle hook. A nonswivel pintle conforming to MS51117 shall be mounted on the rear of the crane carrier.

3.8.4 Tool box. A closed compartment or tool box shall be mounted on the crane in a protected and readily accessible location. The tool box shall be sized to carry all tools and accessories required to operate the crane; all special tools shall be furnished with the crane. The compartment or tool box shall be made of sheet steel of not less than 0.0747 inch thickness. A hasp shall be provided on the tool box to allow locking with a padlock.

3.8.5 Clamshell bucket. When specified (see 6.2), a hydraulically operated clamshell bucket with teeth, shall be provided. Bucket size shall be 1/2 cubic yard for the Size A crane and 5/8 cubic yard for the Size B crane. A combination hose reel and tagline reel shall be attached to control and actuate the bucket.

3.8.6 Maintenance platform. A maintenance platform capable of withstanding a maximum load of 600 pounds, with a minimum floor space of five feet wide and five feet long, shall be provided to accommodate a maintenance crew of two. The platform and sides shall be expanded metal or fiberglass. Each platform shall be provided with a rail, or other structure, around its upper periphery that shall be at least 38 inches above the platform floor, and there shall be a kickplate at least four inches high or its equivalent. The platform shall be attachable to the boom trip to permit a maximum working elevation of at least 47 feet for

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Style A cranes and 70 feet for Style B cranes, a maximum working boom angle of 70° and a minimum working angle of 40° at a boom rotation of 360°. Design shall include a hydraulic cylinder leveler, controls and a level lock in the maintenance platform. Such controls shall render the platform self-leveling with hydraulic level locking. Pins shall permit the platform to be quickly attached to and detached from the crane boom. The maintenance platform shall allow easy access of personnel at ground level and shall include two safety tiedown belts (one belt for each member of the two man maintenance crew). All additional design and performance requirements for the maintenance platform shall be in accordance with ANSI A92.5 excluding platform controls.

3.8.7 Operator Warning Devices. An operator warning device shall be provided. This device shall perform the following functions for any configuration of boom length, parts of line, on rubber or outriggers, over side or rear, and main or auxiliary hoist:

a. Provide an analog or digital reading of the boom angle.

b. Provide an analog or digital reading of length of boom in feet.

c. Provide a drum rotation indicator.

d. Provide a digital reading of the load on hook.

The device shall be powered by the crane's 12 volt electrical system. A two position selector switch, at minimum shall be provided to set the system for either on rubber or on outrigger configuration. The operator warning device shall conform to SAE J375, J376, and J1180.

3.8.8 Level Indicators. The crane shall be equipped with level indicators in accordance with ANSI B30.5.

3.8.9 Mirrors. The crane shall be equipped with two (one on each side) of the manufacturer's standard cab mounted mirrors. They shall be of the combination flat/convex type.

3.9 Finish and protective coatings.

3.9.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808 and/or MIL-T-704, except that commercial (see 6.3.2) components may be prepared in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish paint.

3.9.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defect.

3.9.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the crane and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Forest Green, Color Number 24052 or FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the crane.

3.9.2.2 Desert Sand. When specified (see 6.2), the outer surface of the crane and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the crane.

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3.9.3 Rustproofing. Unless otherwise specified (see 6.2), the crane shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.10 Markings. Unless otherwise specified, markings for cranes painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for cranes painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.10.1 Identification markings. Unless otherwise specified (see 6.2), the crane shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.10.2 Hazard Stripes. Hazard stripes shall be applied to the rear of the chassis, boom point and hook block cheek plates. Stripes shall be manufacturer's standard black, two inches wide, two inches apart, applied at a 45° angle, over specified paint as background.

3.11 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc., shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.11.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the crane in a readily accessible location. The identification shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.11.2 Control markings. All controls, valves, gauges, and indicators used in the operation of the crane shall be identified by nameplates securely installed in clearly visible locations.

3.11.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise, and adequate to enable operation of the crane without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.11.4 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

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a. Side and rear silhouette views of the crane.

b. Horizontal and vertical location of the center of gravity of the crane in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width, and height.

f. Tiedown instructions for air transport in Types C-130, C-141, C-17, and C-5 Aircraft.

g. HQ ASD/ENECA (Air Transportability) certification date and weight certified at.

3.11.5 Loading capacity chart. A permanent chart indicating the safe load capacities for each boom reach and radius for 360° operation with and without outriggers and for pick and carry shall be mounted in clear view of the operator inside the cab. Load capacities shall be based on 85 percent of tipping as determined in accordance with SAE J765.

3.11.6 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operation temperature.

3.12 Dimensions. The overall dimensions of the crane shall not exceed the following dimensions with the boom lowered, retracted and facing forward:

	SIZE A	SIZE B
Width	96 inches	96 inches
	110 inches	136 inches
	28,500 pounds	50,000 pounds

3.13 Noise limits. The exterior sound level produced by the stationary crane shall not exceed 83dB when measured in accordance with SAE J88. In the cab, the interior sound level shall not exceed the octave band pressure levels listed below when measured in accordance with SAE J919:

OCTAVE BAND CENTER FREQUENCY, HERTZ	BAND PRESSURE LEVEL, dB
63	101.5
125	96.0
250	90.5
500	85.0
1,000	79.5
2,000	74.0
4,000	70.0
8,000	70.0

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3.14. Workmanship. The crane, complete with all parts and accessories, shall be fabricated in a thoroughly workmanship manner. Particular attention shall be given to freedom from blemishes, burrs, defects, sharp edges, accuracy of dimensions, radii of fillets and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts and tightness of assembly fasteners; etc. The crane shall be thoroughly cleaned of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the crane shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data shall be recorded at intervals not to exceed 30 minutes, where applicable:

a. Ambient temperature (degrees F).

b. Crane load (pounds).

c. Engine oil pressure (psi).

d. Engine coolant temperature (degree F).

e. Transmission oil temperature (degree F).

f. Fuel level.

g. Engine hours.

4.4 First article testing.

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4.4.1 First article test sample. The first article test sample may be a preproduction item or a production item that meets the requirements of this purchase description. The sample shall be identical to the items procured under terms of this purchase description. Approval of a first article sample shall not constitute a waiver by the Government of its rights under the provision of the contract.

4.4.2 First article test. The first article tests shall consist of all tests listed under 4.6. Unless otherwise approved by the procurring activity, the tests shall be performed in the order listed and with JP-8 fuel in accordance with MIL-T-83133.

4.4.3 First article test report. Upon completion of the first article tests, the contractor shall prepare a test report in accordance with MIL-STD-831 and shall furnish copies of the report to the procuring activity. An authorized Government representative shall countersign all data sheets of the report.

4.5 Acceptance testing. Each crane shall be subjected to the examination of product as described in 4.6.1. Failure to comply with any portion of this purchase description or inability to perform any crane function shall be cause for rejection.

4.6 Test methods.

4.6.1 Examination of product. The crane shall be inspected to determine compliance with the requirements of this purchase description. A thorough operational check shall be made to ensure the proper functioning of the mechanical parts of the crane, accessibility of all controls and levers, interchangeability of removable parts, installation of equipment and safety of operation. The certification necessary to document compliance with all requirements shall be supplied at the first article test and shall be included in the first article test report.

4.6.2 Mobility tests. The crane shall be tested to demonstrate compliance with all requirements of 3.2.2.

4.6.2.1 Travel speed. A fifth wheel shall be attached to the unloaded crane and its maximum forward and reverse speeds shall be determined. An average shall be determined from three runs in opposite directions on a hard, approximately level surface. The crane shall be tested to demonstrate compliance with all requirements of 3.2.2.1.

4.6.2.2 Pick and carry speed. The cranes rated pick and carry load at a 10 foot radius shall be lifted and a fifth wheel attached to the crane. It shall be demonstrated that the crane is capable of a pick and carry speed of at least 2 MPH forward and reverse on a hard, approximately level surface.

4.6.2 3 Cross country operation. The unloaded crane shall be driven over cross country terrain for at least three miles at an average speed not less than 10 MPH to demonstrate compliance with 3.2.2.3.

4.6.2.4 Turning radius. The crane, in four wheel steer, shall be tested to demonstrate compliance with 3.2.2.4. The actual turning radius in each direction shall be measured and recorded.

4.6.3 Winch performance. The crane shall be tested in accordance with SAE J820 to demonstrate compliance with 3.2.3.

4.6.4 Boom rotation speed. With the crane on outriggers, the maximum rated load at the minimum radius for 360° rotation shall be lifted. The time required for a complete rotation in each direction for an unloaded crane shall be determined and recorded. This shall be demonstrated to be in compliance with 3.2.4.

4.6.5 Brake tests. The service brakes and parking brake shall be tested in accordance with SAE J1152 to demonstrate compliance with 3.3.4.1 and 3.3.4.2.

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4.6.6 Crane stability tests. The crane shall be tested to demonstrate compliance with Section 3.1.5 and ANSI B30.5. This test shall be recorded in the format of Figure 1 of SAE J765.

4.6.7 Crane structural test. The crane shall be tested in accordance with SAE J1063. If the crane is a commercial item (see 6.3.2) which has previously satisfied this requirement, a copy of the previous test report, including all data sheets, may be provided in lieu of retesting.

4.6.8 Endurance test. The crane shall lift the maximum rated pick and carry load at a 10 foot radius. The load shall be carried over several surfaces for a distance of not less than eight miles. The average speed for this test shall be 1.5 MPH minimum and the distances on each surface are as follows:

SPEED (MPH)	SMOOTH PAVEMENT	ROUGH TERRAIN	SAND
1.5	2 MILES	4 MILES	2 MILES

The change of surfaces shall occur without stopping. At the conclusion of the durability test, the crane shall be inspected and any damages recorded.

4.6.8.1 Torque converter test. Immediately following the test in 4.6.8, the crane shall be stopped by application of the service brakes and held stationary by the brakes while the transmission is in the lowest gear. The engine shall be run at 90 percent of torque converter stall speed for one minute. The torque converter oil temperature shall not exceed the ambient temperature by more than 125°F.

4.6.9 Electromagnetic interference measurement. The crane and all electrical components shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.1.6.

4.6.10 Final inspection. At the conclusion of testing, the crane shall be thoroughly inspected. Any evidence of a crack, fracture, permanent distortion or a leak shall be cause for rejection.

4.6.11 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.7. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the crane while loaded aboard Types C-130, C-141, C-17, and C-5 Aircraft for size A and Types C-17 and C-5 for size B.

4.6.12 Boom and maintenance platform. Platform shall be shown to meet the requirements of section 3.8.6 and ANSI A92.5.

4.6.13 Noise level measurement. The exterior and interior sound level of the crane shall meet the requirements of section 3.14 and be tested in accordance with SAE J88 and SAE J919.

4.6.14 Stability Demonstration. Verify, with the use of load cells or calibrated weights, that the first article meets the requirements in section 3.1.5.

4.6.15 Operator Warning Device Test. The operator warning device shall be tested to verify compliance with paragraph 3.4.3.1.

4.6.16 Low temperature starting and operation. The complete crane, including all accessories, shall be tested in accordance with Test Method 502.2, Procedure II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). The crane shall be tested for starting and operation at -20°F without external aids (excluding ether injection) and then at -40°F with external aids. A fine mist of water shall be applied to the entire windshield to form a thin film of ice. The defroster shall clear the windshield within one hour and maintain clear vision throughout the remainder of the test. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. The crane shall be exposed to the minimum required cold operational temperature of 3.1.4(a) in a steady-state condition.

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4.7 Refurbishment. After approval of the first article test report, the first article unit shall be restored to like new condition.

5. PACKAGING.

5.1 Packaging. Packaging shall be in accordance with the instructions of the procuring activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 7-1/2 and 15 ton cranes described in this purchase description are intended for lifting and transporting material on smooth surfaces and cross country terrain.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Size required.

c. Nuclear safety, if required (see 3.1.8).

d. Winterization type, Type A (-40°F) or Type C (-20°F) (see 3.3.7, 3.1.4).

e. Cab, if not required (see 3.5.4).

f. Clamshell bucket, if required (see 3.8.5).

g. Exterior finish required, if not polyurethane, dark green. Color Number 24052 of FED-STD-595 (see 3.9.2.1).

h. If polyurethane, Desert Sand, Color Number 30315 of FED-STD-595, is required (see 3.9.2.2).

i. When other than the manufacturer's standard commercial rustproofing is required (see 3.9.3).

j. Marking required (see 3.10.1).

6.3 Definition.

6.3.1 Cross country terrain. Cross country terrain consists of open fields, broken ground, an uneven terrain in general.

6.3.2 Commercial. A manufacturer's standard product, having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is sold in substantial quantities only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes or experimental units, do not meet this requirement. The "general-public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be

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reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475.1 exempts the requirement for a DD Form 1423.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first acticles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Subject term (key word) listing.

Crane

Hydraulic

Wheel Mounted

Custodians:
Air Force - 99

Preparing Activity:
Air Force: 84
Project Number:
3810-F106

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